Parcels R6 & R7, Woodlands, South Marston, Swindon

Vistry Group

Technical Note D-01 – Condition 42

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Contents

Document Control 0
1 Introduction 2
2 Background 3
3 Site Wide Drainage Strategy 5
4 Proposed Drainage Strategy for Parcels R6 & R7 6
5 Summary 7

Appendices

Appendix A – Site Layout
Appendix B – Brookbanks Drainage Strategy
Appendix C – Proposed Surface Water Catchments

1 Introduction

1.1 Rappor UK Limited have been appointed by Vistry Group to assist with the reserved matters design and discharge of conditions for Parcels R6 and R7 of the development scheme.

1.2 The site benefits from outline planning permission (ref. S/OUT/13/1555/EDSN) for a sustainable extension to the east of Swindon and north of the A420 of up to 2,380 dwellings together with a mixed use local centres/areas (including A1 retail up to 1,500 sq metres, services (A2), restaurants, pubs and takeaways (A3, A4, A5), business uses (B1) up to 1,000 sq metres, residential); community uses (D1); sheltered and/or care accommodation (C2/C3); two primary schools; green infrastructure including formal (including playing fields) and informal open space, landscape, biodiversity and amenity space; play space (including NEAPS/LEAPS/MUGAS); changing and sports facilities (including D2) and ancillary retail uses; sustainable drainage systems; highway, cycle and pedestrian routes; car parking; infrastructure (including utilities); engineering works including ground remodelling; demolition, site reclamation and removal of structures; the formation of new accesses from the A420, Old Vicarage Lane and Thornhill Road.

1.3 Subsequently Reserved Matters planning consent (ref S/RES/24/0947) was granted on 04/12/24

1.4 The purpose of this note is to present sufficient information to enable the discharge of planning condition 42 attached to the outline permission.

1.5 Refer to Appendix A for the site layout.

2 Background

2.1 Condition 41 of the outline permission states:

Planning Development shall not begin until a detailed Surface Water Management Strategy for the site, has been submitted to, and approved in writing by the Local Planning Authority. The strategy shall include:

a) Evidence that the proposed flows from the site will be restricted to 4.83l/s/ha for all events up to and including events up to and including the 1% AEP + climate change event (40%);

b) Details of how the drainage strategy has been designed to incorporate SuDS techniques to manage water quantity and maintain water quality as set out in the FRA and adopted policy and best practice guidance including the latest SuDS Manual C753;

c) Details of swales and other overland drainage features to manage the existing surface water flow routes through the site, as highlighted on Plan 1335-DR-05 Rev C (submitted to LPA on 16th July 2020);

d) Detailed drainage plan showing the location of the proposed SuDS and drainage network with exceedance flow routes clearly identified as highlighted on Plan 1335-DR-05 Rev C (submitted to LPA on 16th July 2020), that acknowledges the role of green infrastructure resource to accommodate additional features to augment the detailed design if required);

e) Details of how the strategy shall be maintained and managed after completion;

f) Details of how water quality shall be maintained during and after construction;

g) Detailed drainage calculations for all rainfall events up to and including the 1 in 100 year plus climate change event (40%) to demonstrate that all SuDS features and the drainage network can cater for the critical storm event for its lifetime;

h) The submission of evidence relating to accepted outfalls from the site, particularly from any third party network owners; and

i) Sequencing for implementation.

The scheme shall be implemented in accordance with the approved details and timetable.

Reason: To ensure the surface water management strategy (and future phases of development) will comply with the adopted policy and best guidance and that the development will not increase the risk of flooding elsewhere; in accordance with Paragraph 155 of the National Planning Policy Framework (NPPF) and Policy EN6 of the adopted Swindon Local Plan 2026.

This condition has been discharged under application ref S/COND/22/0711

2.2 Planning condition 42 states:

No development shall take place within each sub-phase until a detailed surface water drainage scheme for that sub-phase has been submitted to and approved in writing by the Local Planning Authority. The scheme shall be in accordance with the surface water

management strategy approved in compliance with Condition 41 and shall include a programme for implementation. The scheme shall include:

a) Detailed Drainage Plan, including proposed site levels, impermeable areas (allowance should be made for runoff from green areas in line with FRA) and exceedance flow routes;

b) Details of the drainage features, including sections and should be designed as set out in the FRA and adopted policy and CIRIA SuDS Manual (C753);

c) Detailed Drainage calculations to support the design for all rainfall events up to and including the 1 in 100 year rainfall event (including an allowance for climate change); Details of Adoption and whole life Maintenance requirements in line with current best practice and include a timetable for the maintenance and for resolving any defects that arise in future; and,

e) Details of Pollution Control / Water Quality.

The scheme shall be implemented in accordance with the approved details.

Reason: To ensure development does not increase the risk of flooding elsewhere in accordance Policy EN6 of the Swindon Borough Local Plan 2026.

2.3 For full Brookbanks surface water drainage report to discharge Condition 41 refer to Brookbanks document 1335/SWDR Rev3.

2.4 Refer to **Appendix B** for the Brookbanks site drainage layout strategy.

3 Site Wide Drainage Strategy

3.1 The strategy provided by Brookbanks in support of the discharge of condition 41 has the following criteria:

a) All parcels including R1 – R22 are to discharge to strategic detention basins and storage swales across the site.

b) At the head of the drainage network, across development parcels, source control measures will be implemented to reduce the amount of run-off being conveyed directly to conveyance systems.

c) The strategy suggests a reduction of 15% in peak flows from residential parcels can be readily achieved

d) Secondary swales (storage swales) will be designed to provide both storage and conveyance functions. The secondary SuDS systems include a number of linear storage and conveyance channels, which are designed to attenuate flows to their maximum available capacity while providing conveyance through the site towards the outfall.

e) In accordance with the 'Surface Water Drainage Strategy' included in the FRA it was agreed that by implementing source control measures, a reduced run-off circa 15% is readily achievable. For the purposes of the primary Drainage System appraisals and inputs to the secondary system, flows from the MD Source Control model assessment, at the sub-catchment level, have therefore been modelled as equivalent to a reduced contributing catchment of 15%.

f) Parcels R6 and R7 discharge to Detention Basin A. Drainage catchment details are as below:

Parcel	Use	Area (Ha)	Hard Area (Ha)	Hard Area (Ha) With 10% UC	Increase due to 10% UC (Ha)		Hard area (Ha) Reduced by 15%	Hard Area (Ha) - Reduced by 15% with UC increase		Difference Reduced vs Original (Ha)
R6	Residential	3.33	1.83	1.94	0.11		1.56	1.67		0.27
R7	Residential	1.06	0.58	0.62	0.03		0.5	0.53		0.09
		4.39	**2.41**	**2.56**	**0.14**		**2.06**	**2.2**		**0.36**

3.2 As per the Brookbanks strategy (summarised above), a reduction of 15% of the run-off is to be provided and attenuated on site. The following options were proposed by Brookbanks, however, we (Rappor) have provided a response to why these are or are not feasible:

a) Permeable Paving – Feasible
Permeable paving can be utilised on the development where space allows (i.e. beneath shared surfaces where block paving is typically used). Permeable paving is a feasible option as any attenuation can be achieved within the sub-base layer.

b) Filter Strips / Ditches / Swales – Not Feasible
Filter strips, ditches and swales are not feasible to be used due to the space required for them to gain the required attenuation volumes. Additionally, the site layout only provides enough space for these along the spine road between parcels R6 and R7, therefore unable to contribute to the required 15% reduction.

3.3 Therefore the only solution that can be implemented from the above is the use of permeable paving. The following section summarises the use of permeable paving and how the 15% reduction has been achieved.

4 Proposed Drainage Strategy for Parcels R6 & R7

4.1 The Brookbanks strategy proposes that parcel R7 discharges all to the same location (i.e. the strategic infrastructure). However, this was assumed without a full understanding of levels for the site.

4.2 The dwellings on the northeastern side of the parcel front onto Old Vicarage Lane and as a result are too low to drain as per the strategic strategy, back into the strategic drainage located in the spine road that separates parcels R6 and R7. Therefore, a proportion of parcel R7 has been designed to discharge into the ditch adjacent to Old Vicarage Lane at greenfield run-off rates (QBAR). The existing run-off flows towards this ditch, therefore discharging at QBAR does not have an effect on the existing surface water catchments. Attenuation for this section of R7 is provided in permeable paving, thereby providing an element of source control also.

4.3 For Parcel R6 and the remaining areas of R7 source control is used in the form of permeable paving.

4.4 The areas are summarised in the below table, please refer to **Appendix C** for the corresponding catchment plan

Parcel	Catchment	Hard Area (ha)	Hard Area (Ha) With 10% UC	Hard Area (Ha) - Reduced by 15% with UC increase	Difference Reduced vs Original (Ha)		Actual Area Reduction (Ha)	
R6	Red	0.629	0.649	0.555	0.094		0.052	From Private Shared Space
R6	Blue	0.276	0.283	0.242	0.041		0.036	From Private Shared Space
R6	Yellow	0.818	0.846	0.723	0.123		0.038	From Private Shared Space
R6 / R7	Cyan	0.472	0.494	0.423	0.071		0.121	From Private Shared Space
R7	Green	0.176	0.184	0.157	0.026		0.184	From Whole Catchment
		2.370	**2.455**	**2.100**	**0.355**		**0.430**	
					Total Percentage Difference (Actual Area Reduction vs Hard Area with 10% UC)		**17.5%**	

4.5 The total impermeable area for parcels R6 and R7 (including Urban Creep) has been calculated as 2.455ha.

4.6 Utilising the private shared spaces for permeable paving, and excluding the green catchment from parcel R7 now draining to the existing ditch, the total area reduction is 0.430ha.

4.7 This provides a percentage betterment of 17.5% and therefore exceeds the 15% requirement.

5 Summary

5.1 Based on the information presented in this Technical Note, it can be concluded that the parcels R6 and R7 satisfy the requirements set out in the Brookbanks FRA, and in turn satisfy the relevant parts of condition 42.

5.2 Using permeable paving in private shared spaces, and re-directing a proportion of Parcel R7 into the existing adjacent ditch, the required 15% reduction in run-off area is exceeded.

Appendix A – Site Layout